Guide to Econometrics and Time Series

Cross-section, panel data, time series,
volatility and machine learning

Study material with original explanations,
numerically checked experiments and worked exercises

Reference books for further reading: Wooldridge, Goldberger, Stachurski, Hanck et al., Hamilton, Lütkepohl, Cryer and Chan, Cowpertwait and Metcalfe, Mills, Tsay, Montgomery et al., Hang Chan, Hastie et al., Faloutsos et al., Faouzi and Janati.

Contents

Before you start **1**

I Foundations **3**

1 Conditional expectation and linear projection **4**

1.1 What conditional expectation is for 4

1.2 The error form 4

1.3 Partial effects, elasticities and semi-elasticities 4

1.4 Linear projection: what OLS really estimates 5

2 Asymptotics in plain language **6**

2.1 Why look at large samples 6

2.2 Consistency is not unbiasedness 6

2.3 The pattern behind all standard errors 6

II Regression and identification **8**

3 OLS: what it estimates, when it works and how to compute standard errors **9**

3.1 The model and the minimal assumptions 9

3.2 Consistency in two lines 9

3.3 Standard errors: usual versus robust 9

4 Omitted variables and measurement error **11**

4.1 Omitted variable 11

4.2 Proxy variable 11

4.3 Classical measurement error 11

5 Instrumental variables and 2SLS **13**

5.1 The idea 13

5.2 The simple IV estimator 13

5.3 2SLS (two-stage least squares) . . . 13
5.4 Weak instruments . . . 14

III Panel data **15**

6 Panel data **16**

6.1 The unobserved effects model . . . 16
6.2 The main estimators . . . 16
6.3 Fixed or random effects? The Hausman test and the Mundlak alternative . . . 17

IV Time series **18**

7 Time series: stationarity and autocorrelation **19**

7.1 Why time series need their own tools . . . 19
7.2 Weak stationarity . . . 19
7.3 Four series you need to know . . . 19
7.4 AR(2) and the stationarity condition . . . 20
7.5 Signatures in the ACF and PACF . . . 20

8 ARMA, ARIMA and the Box–Jenkins methodology **22**

8.1 The ARMA model and the lag operator . . . 22
8.2 ARIMA: what the letter "I" does . . . 22
8.3 The Box–Jenkins cycle in four steps . . . 22
8.4 Seasonality . . . 23

9 Forecasting **24**

9.1 The best predictor . . . 24
9.2 Random walk and ARIMA . . . 24
9.3 How to evaluate forecasts . . . 25

10 Unit roots, spurious regression and cointegration **26**

10.1 Deterministic versus stochastic trend . . . 26
10.2 The augmented Dickey–Fuller (ADF) test . . . 26

10.3 Spurious regression 27
10.4 Cointegration 27

11 Vector autoregressions (VAR) 29

11.1 The model 29
11.2 Choosing the number of lags 29
11.3 Granger causality 29
11.4 Impulse response functions and variance decomposition 30

12 Volatility: ARCH and GARCH 31

12.1 The problem 31
12.2 ARCH(1) and GARCH(1,1) 31

13 Machine learning for time series 33

13.1 What changes when data have an order 33
13.1.1 Cross-validation that respects time 33
13.1.2 Trees do not extrapolate 33
13.1.3 Building the explanatory variables 34
13.2 Two useful families 34

V Quick reference 35

14 Pitfalls and conventions cheat sheet 36

Before you start

This guide was written from scratch, with original explanations, to help you learn cross-section econometrics, panel data and time series. It is *not* a translation of any of the books. The books are the source for deeper reading. This material is the didactic path that leads you to them.

How the guide is organized

Each chapter follows the same order: the idea in plain language, the essential mathematics, a numerical experiment that confirms the theory, the most common pitfalls, and exercises with solutions. The colored boxes have fixed roles:

Intuition

Explains the *why* before the formula.

Pitfall

Frequent mistakes, including some that appear in books and software manuals.

Checked by simulation

A result I ran on the computer (Python) to make sure the statement is correct. The numbers come from a fixed seed, so they vary slightly if you repeat the experiment.

Map of the books: where to go deeper

Topic	Where to read more
Background probability and linear algebra	Stachurski (*Econometric Theory*, lecture notes), parts I and II; Goldberger
Regression, OLS, IV	Wooldridge (ch. 4 and 5 of the 1st edition); Hanck et al. (ch. 4 to 9 and 12)
Panel data	Wooldridge (ch. 10 and 11); Hanck et al. (ch. 10)
ARMA, ARIMA, Box–Jenkins	Cryer and Chan (ch. 4 to 9); Mills (ch. 3, 4 and 7); Hamilton (ch. 3 and 4)
Unit roots and cointegration	Hamilton (ch. 17 to 19); Mills (ch. 14); Hanck et al. (ch. 14 and 16)
VAR	Lütkepohl (*New Introduction to Multiple Time Series Analysis*); Hamilton (ch. 11); Mills (ch. 13)
Volatility (ARCH/GARCH)	Tsay (*Analysis of Financial Time Series*); Mills (ch. 10); Hamilton (ch. 21)
Machine learning	Hastie, Tibshirani and Friedman (*Elements of Statistical Learning*); Faloutsos et al. tutorial on large-scale forecasting; Faouzi and Janati (`pyts` package)

Two remarks about the books

Wooldridge edition. The file you sent is the 1st edition of *Econometric Analysis of Cross Section and Panel Data* (2002, 2004 printing). A 2nd edition appeared in 2010 and is considerably longer. The concepts I use here (OLS, IV, fixed and random effects) are in both, but chapter numbering and some advanced topics change. For current applied work, the 2nd edition is worth having.

About "correcting the books' errors". I did not audit thousands of pages looking for errata, and I will not pretend I did. What I did was (i) check by simulation every formula and number that appears in this guide and (ii) highlight, in the *Pitfall* boxes, the points where books disagree with each other or where a mistake is common. The most important one is the sign of the moving-average coefficient, which changes from book to book (chapter 7). I also did not read the books cover to cover: I used the tables of contents to build the map above and checked directly in Cryer–Chan and in Mills the sign convention of the MA. If you find something that does not match your book, trust the simulation and the derivation, not my text.

Notation

Vectors and matrices are in bold when useful. $\mathbb{E}(\cdot)$ is expectation, Var variance, Cov covariance, plim probability limit. $\hat{\beta}$ is an estimator and β the true parameter. In time series, B is the lag operator: $By_t = y_{t-1}$.

Part I

Foundations

Chapter 1

Conditional expectation and linear projection

1.1 What conditional expectation is for

Almost all of econometrics tries to answer: *given what I know about x, what is the typical value of y?* The mathematical answer is the conditional expectation $m(x) = \mathbb{E}(y \mid x)$.

Intuition

Imagine splitting people by years of schooling (x) and computing the average wage of each group (y). The list of those group averages is $m(x)$. It says nothing about cause. It only says what the average wage is for people with that schooling.

Result 1.1 (Conditional expectation is the best predictor). *For any function $g(x)$,*

$$\mathbb{E}[(y - g(x))^2] \;=\; \mathbb{E}[(y - m(x))^2] \;+\; \mathbb{E}[(m(x) - g(x))^2] \;\geq\; \mathbb{E}[(y - m(x))^2].$$

Hence $m(x)$ minimizes the mean squared error among all functions of x.

Why? Add and subtract $m(x)$ inside the square. The cross term has zero expectation because $\mathbb{E}(y - m(x) \mid x) = 0$.

1.2 The error form

Define $u = y - m(x)$. Then $y = m(x) + u$ with $\mathbb{E}(u \mid x) = 0$. Two consequences used all the time:

- $\mathbb{E}(u) = 0$ (by the law of iterated expectations, $\mathbb{E}(u) = \mathbb{E}[\mathbb{E}(u \mid x)]$);
- $\mathrm{Cov}(u, h(x)) = 0$ for any function h (the error has no relation with x, neither linear nor nonlinear).

1.3 Partial effects, elasticities and semi-elasticities

If $m(x_1, x_2)$ is the conditional expectation, the partial effect of x_1 is $\partial m / \partial x_1$. With variables in logs:

Model	Coefficient β_1 measures	Name
$y = \beta_0 + \beta_1 x_1$	change in y per unit of x_1	marginal effect
$\log y = \beta_0 + \beta_1 x_1$	$\approx 100\beta_1\%$ change in y per unit of x_1	semi-elasticity
$\log y = \beta_0 + \beta_1 \log x_1$	% change in y per 1% in x_1	elasticity

If the model has an interaction or a quadratic term, the effect of x_1 changes with the values of the variables. In that case report the **average partial effect** (APE): the average of the partial effect over the distribution of x.

1.4 Linear projection: what OLS really estimates

$m(x)$ is not always linear. Even so, there is always a best linear approximation in the mean squared error sense:

$$L(y \mid 1, x) = \beta_0 + x\beta, \qquad \beta = \left[\mathbb{E}(x'x)\right]^{-1}\mathbb{E}(x'y) \quad \text{(with } x \text{ including the constant)}.$$

Intuition

The linear projection is the line (or hyperplane) that best *summarizes* the cloud of points, even if the cloud is curved. OLS estimates exactly this line. Only when $m(x)$ really is linear does this line coincide with the conditional expectation.

Pitfall: "linear" is not "causal"

The projection coefficient is always well defined. That does not make it a causal effect. For β to have a causal reading, you need an extra argument about why the other factors that affect y do not move together with x (chapters 3 to 5).

Exercise 1.1. Let $x \sim N(0, 1)$ and $y = x^2$ (no noise). What is the slope of the linear projection of y on x? What does this say about dependence between y and x?

Solution. $\mathrm{Cov}(x, x^2) = \mathbb{E}(x^3) - \mathbb{E}(x)\mathbb{E}(x^2) = 0$, because odd moments of a centered normal are zero. So the slope is zero and the projection is the constant $\mathbb{E}(y) = 1$. Yet y is an exact function of x. A zero correlation (or slope) does **not** mean independence. This is the shortest example of why the line can hide a strong relationship.

Exercise 1.2. Show that if $\mathbb{E}(u \mid x) = 0$, then $\mathrm{Cov}(u, x^2) = 0$.

Solution. $\mathrm{Cov}(u, x^2) = \mathbb{E}(ux^2) - \mathbb{E}(u)\mathbb{E}(x^2)$. By the law of iterated expectations, $\mathbb{E}(ux^2) = \mathbb{E}[x^2\mathbb{E}(u \mid x)] = 0$, and $\mathbb{E}(u) = 0$. Therefore the covariance is zero.

Chapter 2

Asymptotics in plain language

2.1 Why look at large samples

With small samples, we only get exact properties if we make strong assumptions, such as normal errors. With large samples, the mathematics gets lighter and the assumptions more realistic. Four tools are enough for the whole course.

Tool	What it says
Law of large numbers (LLN)	The sample mean $\bar{x}_n$ converges in probability to $\mathbb{E}(x)$.
Central limit theorem (CLT)	$\sqrt{n}(\bar{x}_n - \mu)$ converges in distribution to $N(0, \sigma^2)$, whatever the distribution of x (with finite variance).
Slutsky's theorem	If $a_n \to_p a$ and $b_n \to_d b$, then $a_n b_n \to_d ab$ and $a_n + b_n \to_d a + b$. It lets us replace unknown parameters by consistent estimates.
Delta method	If $\sqrt{n}(\hat{\theta} - \theta) \to_d N(0, V)$, then $\sqrt{n}(g(\hat{\theta}) - g(\theta)) \to_d N(0, g'(\theta)^2 V)$.

2.2 Consistency is not unbiasedness

- **Unbiased**: $\mathbb{E}(\hat{\theta}) = \theta$ for a fixed sample size.
- **Consistent**: $\hat{\theta} \to_p \theta$ as $n \to \infty$.

An estimator can have one property and not the other. The mean of only the first two observations is unbiased and not consistent. The estimator $\frac{1}{n}\sum(x_i - \bar{x})^2$ of the variance is biased in finite samples and consistent. In applied econometrics, consistency is usually what matters, because almost no useful estimator is exactly unbiased.

2.3 The pattern behind all standard errors

Most estimators we will see satisfy

$$\sqrt{n}(\hat{\beta} - \beta) \ \to_d \ N(0, \ A^{-1}BA^{-1}),$$

where A is the "curvature matrix" and B is the variance of the score. This "sandwich" form reappears in OLS with robust errors (ch. 3), in IV (ch. 5) and in panel data with clustered errors (ch. 6). When homoskedasticity holds, B is proportional to A and the expression simplifies to $\sigma^2 A^{-1}$.

Exercise 2.1. Let x_i be i.i.d. with mean $\mu = 2$ and variance $\sigma^2 = 9$. What is the approximate standard error of $\bar{x}$ with $n = 400$, and what is a 95% interval if $\bar{x} = 2.3$?

Solution. se $= \sigma/\sqrt{n} = 3/20 = 0.15$. Approximate interval: $2.3 \pm 1.96 \times 0.15 = [2.006;\ 2.594]$. If σ were unknown, we would replace it with a consistent estimator (Slutsky) and the result would be the same in large samples.

Part II

Regression and identification

Chapter 3

OLS: what it estimates, when it works and how to compute standard errors

3.1 The model and the minimal assumptions

Consider $y = x\beta + u$, where $x = (1, x_1, \dots, x_k)$ is a row vector. The ordinary least squares estimator is $\hat{\beta} = (X'X)^{-1}X'y$.

The four assumptions for consistency and asymptotic normality, in the random-sample version:

1. $y = x\beta + u$ (the model is linear in the parameters);
2. the data are a random sample (independent and identically distributed observations);
3. $\mathbb{E}(x'u) = 0$ (the regressors are uncorrelated with the error);
4. $\mathbb{E}(x'x)$ has full rank (no perfect collinearity).

Pitfall: the conditional assumption is stronger than necessary

Many classic texts require $\mathbb{E}(u \mid x) = 0$ or treat x as fixed. For *consistency*, $\mathbb{E}(x'u) = 0$ is enough, which is the assumption that the linear projection is correct. $\mathbb{E}(u \mid x) = 0$ is only needed if you want to interpret $x\beta$ as the conditional expectation, or if you want finite-sample properties (exact unbiasedness). Wooldridge insists on this distinction and it is worth remembering.

3.2 Consistency in two lines

$$\hat{\beta} = \beta + \left(\frac{1}{n}\sum x_i'x_i\right)^{-1}\left(\frac{1}{n}\sum x_i'u_i\right) \;\to_p\; \beta + \left[\mathbb{E}(x'x)\right]^{-1}\mathbb{E}(x'u) = \beta.$$

The second factor goes to zero by the LLN *if* $\mathbb{E}(x'u) = 0$. All the rest of the course (omitted variables, measurement error, IV) is a story about what happens when this condition fails.

3.3 Standard errors: usual versus robust

- **Usual** (homoskedastic): $\widehat{\mathrm{Var}}(\hat{\beta}) = \hat{\sigma}^2(X'X)^{-1}$. Valid if $\mathrm{Var}(u \mid x)$ is constant.
- **Heteroskedasticity-robust** (White/Eicker–Huber): $\widehat{\mathrm{Var}}(\hat{\beta}) = (X'X)^{-1}\left(\sum \hat{u}_i^2 x_i'x_i\right)(X'X)^{-1}$. Valid even when the error variance depends on x.

Checked by simulation: the usual standard error understates

$n = 5000$, $y = 1 + 2x + |x|\,\varepsilon$, with $\varepsilon \sim N(0,1)$ (the error variance grows with $|x|$). The coefficient of x is estimated without bias, but the usual standard error is 0.0143 and the robust one is 0.0248. The usual one is about 42% smaller, and would produce inflated t tests and intervals that are too narrow.

Why Gauss–Markov matters less than it seems

The Gauss–Markov theorem says OLS is the best linear unbiased estimator *if* there is homoskedasticity. With heteroskedasticity it remains consistent; it just stops being the most efficient, and the usual standard error stops being valid. So, in practice, the modern habit is: estimate by OLS and report robust errors.

Frequent pitfalls

- **A high R^2 does not validate causality.** It measures fit, not identification.
- **Testing for heteroskedasticity before choosing the standard error** is unnecessary: the robust one is valid in both cases.
- **Time series or panels**: the heteroskedasticity-robust standard error does *not* correct serial correlation. Use autocorrelation-robust (HAC) or clustered errors.

Exercise 3.1. A model $y = \beta_0 + \beta_1 x + u$ was estimated with $n = 100$, $\hat{\beta}_1 = 0.8$ and robust standard error 0.3. Test $H_0 : \beta_1 = 0$ at 5%.

Solution. $t = 0.8/0.3 \approx 2.67 > 1.96$. Reject H_0 at 5%. 95% interval: $0.8 \pm 1.96 \times 0.3 = [0.21;\ 1.39]$.

Chapter 4

Omitted variables and measurement error

4.1 Omitted variable

Suppose the true model is $y = \beta_0 + \beta_1 x + \beta_2 z + u$, but you do not observe z and regress y on x. Let δ_1 be the slope of the regression of z on x. Then

$$\operatorname{plim} \hat{\beta}_1^{\text{short}} = \beta_1 + \beta_2\, \delta_1, \qquad \delta_1 = \frac{\operatorname{Cov}(x, z)}{\operatorname{Var}(x)}.$$

The bias is the product of two factors: the effect of z on y (β_2) and the association between z and x (δ_1). If either is zero, there is no bias.

Checked by simulation

$x = 0.6z + \epsilon$, with $z, \epsilon \sim N(0,1)$ independent, and $y = 1 + 2x + 1.5z + u$. Theory: $\delta_1 = 0.6/1.36 = 0.441$, so $\operatorname{plim} \hat{\beta}_1^{\text{short}} = 2 + 1.5 \times 0.441 = 2.662$. With 2 million observations, the short regression gives 2.662. Including z, the coefficient returns to 2.01 (at $n = 5000$).

Intuition

Without z in the equation, x "hitches a ride" on the effect of z on y because the two move together. The stronger the ride, the larger the bias. The sign of the bias is the sign of $\beta_2 \delta_1$: if both are positive, OLS overstates.

4.2 Proxy variable

If z is not observed but you have a proxy w (for example, a test score for ability), including w can reduce or eliminate the bias. It works if (i) w is *redundant* in the equation for y when z is present and (ii) z is, conditionally, linearly related to w without depending on x. If (ii) fails, the proxy can even make things worse, since an imperfect control leaves residual bias.

4.3 Classical measurement error

There are two cases with very different consequences.

- **Error in the dependent variable**, $y = y^* + e$ with e uncorrelated with x: OLS remains consistent and only loses precision.

- **Error in an explanatory variable**, $x = x^* + v$ with v independent of x^* and of u: OLS is *attenuated* toward zero:

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 \cdot \frac{\sigma^2_{x^*}}{\sigma^2_{x^*} + \sigma^2_v}.$$

  The factor in parentheses is the "reliability ratio", always between 0 and 1.

Checked by simulation

$x^* \sim N(0,1)$, $y = 1 + 2x^* + u$, and $x = x^* + v$ with $v \sim N(0,1)$. The reliability ratio is $1/(1+1) = 0.5$, so OLS should converge to $2 \times 0.5 = 1.0$. Result: 0.995.

Pitfall

Attenuation holds for the regressor with error. If there are *other* regressors correlated with it, the bias can contaminate the remaining coefficients in directions that are not obvious. And if the measurement error is correlated with the true value ("nonclassical error"), not even the sign of the attenuation is guaranteed.

Exercise 4.1. In a model where $\beta_1 = 0.5$, $\beta_2 = 2$, $\operatorname{Var}(x) = 4$ and $\operatorname{Cov}(x, z) = 1$, what is the probability limit of OLS of y on x if z is omitted?

Solution. $\delta_1 = 1/4 = 0.25$. $\operatorname{plim} \hat{\beta}_1 = 0.5 + 2 \times 0.25 = 1.0$. OLS doubles the true effect.

Exercise 4.2. An economist measures income with error and obtains $\hat{\beta} = 0.30$. She knows the reliability of the measure is 0.6. What would a corrected estimate of the attenuation be?

Solution. $\hat{\beta}/0.6 = 0.30/0.6 = 0.50$, assuming classical error and a single regressor. In practice, the reliability has to come from outside (repeated data, validation, or an instrument).

Chapter 5

Instrumental variables and 2SLS

5.1 The idea

When x is correlated with u (endogeneity), OLS is inconsistent. An **instrument** z is a variable that moves x but affects y only *through* x. Two conditions:

1. **Relevance**: $\text{Cov}(z, x) \neq 0$ (conditional on the other exogenous regressors). It is testable.
2. **Exogeneity**: $\text{Cov}(z, u) = 0$. In a model with exactly one instrument per endogenous variable, it is **not testable**. It is an economic argument, not a statistical one.

5.2 The simple IV estimator

With one regressor and one instrument,

$$\hat{\beta}_{IV} = \frac{\sum(z_i - \bar{z})(y_i - \bar{y})}{\sum(z_i - \bar{z})(x_i - \bar{x})} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)}.$$

Intuition

The numerator measures how much y changes when z changes. The denominator measures how much x changes when z changes. The ratio is the effect of x on y extracted only from the variation induced by z. This is the "Wald" version of IV.

Exercise 5.1 (Hand calculation)**.** $z = (0, 0, 1, 1, 1, 0)$, $x = (1, 2, 2, 4, 5, 3)$, $y = (2, 3, 6, 8, 9, 3)$. Compute $\hat{\beta}_{IV}$ and compare with OLS.

Solution. Means: $\bar{z} = 0.5$, $\bar{x} = 2.833$, $\bar{y} = 5.167$. $\sum(z - \bar{z})(y - \bar{y}) = 7.5$ and $\sum(z - \bar{z})(x - \bar{x}) = 2.5$, so $\hat{\beta}_{IV} = 7.5/2.5 = 3.0$. OLS gives 1.677. The difference between the two is large, which would be a sign of endogeneity if z were a good instrument (with $n = 6$ this is just an exercise in arithmetic).

5.3 2SLS (two-stage least squares)

With several instruments Z (which include the exogenous regressors), the estimator is

$$\hat{\beta}_{2SLS} = (X' P_Z X)^{-1} X' P_Z y, \qquad P_Z = Z(Z'Z)^{-1} Z'.$$

In words: in the first stage, each endogenous variable is regressed on the instruments; in the second, the fitted value is used in place of the original variable.

Checked by simulation

Endogeneity. $x = 0.8z + 0.7u + \epsilon$, $y = 1 + 2x + u$. OLS converges to a wrong value, 2.33. 2SLS recovers 2.02 (true value: 2). The first-stage F statistic was ≈ 2090.
Matrix formula. With two instruments ($n = 3000$), I computed $(X'P_Z X)^{-1}X'P_Z y$ by hand and with the `linearmodels` package. Both gave exactly (0.97599; 1.97238) for (constant, slope).

5.4 Weak instruments

If $\mathrm{Cov}(z, x)$ is close to zero, the denominator of IV gets close to zero and the estimator becomes unstable and biased toward OLS.

Weak instrument

Same design, but $x = 0.02z + 0.7u + \epsilon$. Here OLS converges to ≈ 2.47 (larger bias, since x has less variation of its own besides u). In a simulation with $n = 5000$, 2SLS gave 2.30: far from the true value (2) and pulled toward OLS. With an instrument this weak, the result of a single sample is very unstable. The instrument exists, but it does not help.

Classic IV pitfalls

- **Running the 2nd stage "by hand" and using the second OLS standard error**: this is wrong. The residuals used must be computed with the *original* x, not the fitted value. Use a 2SLS command, which makes the correction.
- **The "first-stage F $>$ 10" rule** is a minimum floor, not a guarantee. With many weak instruments, bias persists.
- **Overidentification test** (Sargan/Hansen) only tests whether the instruments agree with each other. If all are invalid in the same way, it does not notice.
- **IV estimates a local effect**: with heterogeneous effects, it estimates the average effect among those whose x changes because of z, not the effect for the whole population.

Exercise 5.2. Why is it inadequate to use as an instrument a variable that affects y directly, even if it is strongly correlated with x?

Solution. Because it violates exogeneity: part of the correlation between z and y does not pass through x, and the ratio $\mathrm{Cov}(z, y)/\mathrm{Cov}(z, x)$ now includes that direct channel. The estimator converges to $\beta + \mathrm{Cov}(z, u)/\mathrm{Cov}(z, x)$, and the numerator is no longer zero.

Part III

Panel data

Chapter 6

Panel data

6.1 The unobserved effects model

We have N units (people, firms, municipalities) observed over T periods:

$$y_{it} = x_{it}\beta + c_i + u_{it}.$$

c_i is the unobserved effect, constant over time (for example, ability or firm culture). The central question is: **is c_i correlated with x_{it}?**

6.2 The main estimators

Estimator	What it does	Consistent if
Pooled OLS (POLS)	Ignores c_i and runs OLS on everything	$\mathbb{E}(x'_{it}(c_i + u_{it})) = 0$, that is, c_i uncorrelated with x
Fixed effects (within)	Subtracts each unit's mean, eliminating c_i	Strict exogeneity: u_{it} uncorrelated with x_{is} in *all* periods
First differences	Regresses Δy_{it} on Δx_{it}	Same; with $T = 2$ it coincides with fixed effects
Random effects (GLS)	Subtracts a fraction θ of the unit mean	c_i uncorrelated with x_{it} (strong assumption)

Intuition

Fixed effects uses only the variation *within* each unit over time. Everything that is fixed within the unit (sex, place of birth, culture) is automatically neutralized, observed or not. The price: you lose the ability to estimate the effect of any variable that does not change over time, and you lose variation if x changes little.

Checked by simulation: correlated effect

$N = 500$, $T = 6$, $y_{it} = 1 + 1.0\,x_{it} + c_i + u_{it}$, with $x_{it} = 0.7c_i + \text{noise}$ (that is, c_i and x are correlated). The true effect is 1.0.

	Pooled OLS	Random effects	Fixed effects	First diff.
Estimate	1.47	1.27	0.98	1.00

Pooled OLS and random effects are biased because they assume c_i is not correlated with x. Fixed effects and first differences recover the true value.

6.3 Fixed or random effects? The Hausman test and the Mundlak alternative

The Hausman test compares fixed and random effects. A more flexible version, which in practice is more used, is the **Mundlak approach**: include the time average $\bar{x}_i$ as an additional regressor and test whether the coefficient of $\bar{x}_i$ is zero. If it is zero, random effects is reasonable. The coefficient of $\bar{x}_i$ measures the difference between the "between-unit" and "within-unit" effects.

Checked by simulation

With c_i correlated with x, the coefficient of $\bar{x}_i$ was 1.06, with a p-value practically zero: random effects is rejected. With c_i independent of x, the coefficient was 0.16 (p-value 0.26): not rejected. The value ≈ 1.06 matches the theory: the difference between the "between" effect (≈ 2.07) and the "within" effect (1).

Pitfalls in panels

- **The classic Hausman test under robust/clustered errors** no longer has the expected distribution. Prefer the regression version (Mundlak) with clustered errors.
- **Always cluster the standard errors by unit.** c_i generates serial correlation within the unit, even when u_{it} is independent. The usual pooled OLS standard error is typically too small.
- **Strict exogeneity is strong.** Lagged dependent variables as regressors, or *feedback* (y_{it} affects $x_{i,t+1}$), violate it. In those cases, dynamic panel methods (GMM) are needed, which go beyond this chapter.
- **Fixed effects do not solve endogeneity that varies over time.**

Exercise 6.1. With $T = 2$, show that the fixed effects estimator equals the first-differences estimator.

Solution. With $T = 2$, the deviation from the mean is $\ddot{y}_{i1} = (y_{i1} - y_{i2})/2 = -\Delta y_i/2$ and $\ddot{y}_{i2} = \Delta y_i/2$ (and the same holds for x). Then $\sum_t \ddot{x}_{it}\ddot{y}_{it} = \frac{1}{2}\Delta x_i \Delta y_i$ and $\sum_t \ddot{x}_{it}^2 = \frac{1}{2}(\Delta x_i)^2$. The ratio of the sums over i is $\sum \Delta x_i \Delta y_i / \sum (\Delta x_i)^2$, which is exactly OLS in first differences.

Exercise 6.2. Why can the effect of "years of schooling" not be estimated by fixed effects in a sample of adults who have already finished school?

Solution. Because schooling does not vary within each person over the panel. After subtracting the person's mean, the regressor becomes zero in all periods, and the coefficient is not identified.

Part IV

Time series

Chapter 7

Time series: stationarity and autocorrelation

7.1 Why time series need their own tools

In cross-section data, independent observations are reasonable. In a series, today's value depends on yesterday's. This changes everything: the same series is our only sample, and we can only learn from it if its behavior is, in some sense, stable over time. That stability has a name: **stationarity**.

7.2 Weak stationarity

A series $\{y_t\}$ is **weakly stationary** (or covariance stationary) if

1. $\mathbb{E}(y_t) = \mu$ does not depend on t;
2. $\mathrm{Var}(y_t) = \gamma_0$ does not depend on t;
3. $\mathrm{Cov}(y_t, y_{t-k}) = \gamma_k$ depends only on the lag k, not on t.

The **autocorrelation function** (ACF) is $\rho_k = \gamma_k/\gamma_0$. The **partial ACF** (PACF) at k is the correlation between y_t and y_{t-k} after removing the effect of the intermediate lags.

ACF versus PACF

The ACF mixes direct and indirect effects: if today depends on yesterday and yesterday on the day before, today and two days ago are correlated even without a direct link. The PACF tries to isolate only the direct link.

7.3 Four series you need to know

Series	Definition	Properties
White noise	$y_t = \varepsilon_t$, mean 0, variance σ^2, uncorrelated	$\rho_k = 0$ for $k \geq 1$. Building block of all models.
Random walk	$y_t = y_{t-1} + \varepsilon_t$	**Not** stationary: the variance $t\sigma^2$ grows. Shocks are permanent.
AR(1)	$y_t = \phi y_{t-1} + \varepsilon_t$	Stationary if $\lvert\phi\rvert < 1$; $\rho_k = \phi^k$ (geometric decay).
MA(1)	$y_t = \varepsilon_t + \theta\varepsilon_{t-1}$	Always stationary; $\rho_1 = \theta/(1+\theta^2)$ and $\rho_k = 0$ for $k \geq 2$.

Pitfall: the sign of the MA changes from book to book

This guide uses $y_t = \varepsilon_t + \theta\varepsilon_{t-1}$ (**positive** sign), like Hamilton, R's `arima` and `statsmodels`. Cryer and Chan and Mills use the **negative** sign, $y_t = \varepsilon_t - \theta\varepsilon_{t-1}$, inherited from Box and Jenkins. With the negative convention, $\rho_1 = -\theta/(1+\theta^2)$. If you copy a formula from a book and a coefficient from a software package without checking the convention, the sign of θ comes out reversed. When reading software output, always confirm which convention it uses.

Checked: MA(1) with coefficient 0.5

The formula gives $\rho_1 = 0.5/1.25 = 0.4$. The theoretical autocorrelation computed by `statsmodels` is 0.4. Note also that $\theta = 2$ gives $2/5 = 0.4$: **two values of θ (one and its inverse) generate the same ACF**. That is why invertibility, $|\theta| < 1$, is required to identify the model. And the maximum of $|\rho_1|$ in an MA(1) is 0.5.

7.4 AR(2) and the stationarity condition

For $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$, the three conditions below are equivalent to "roots of the polynomial outside the unit circle" (the so-called stationarity triangle):

$$\phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1, \qquad |\phi_2| < 1.$$

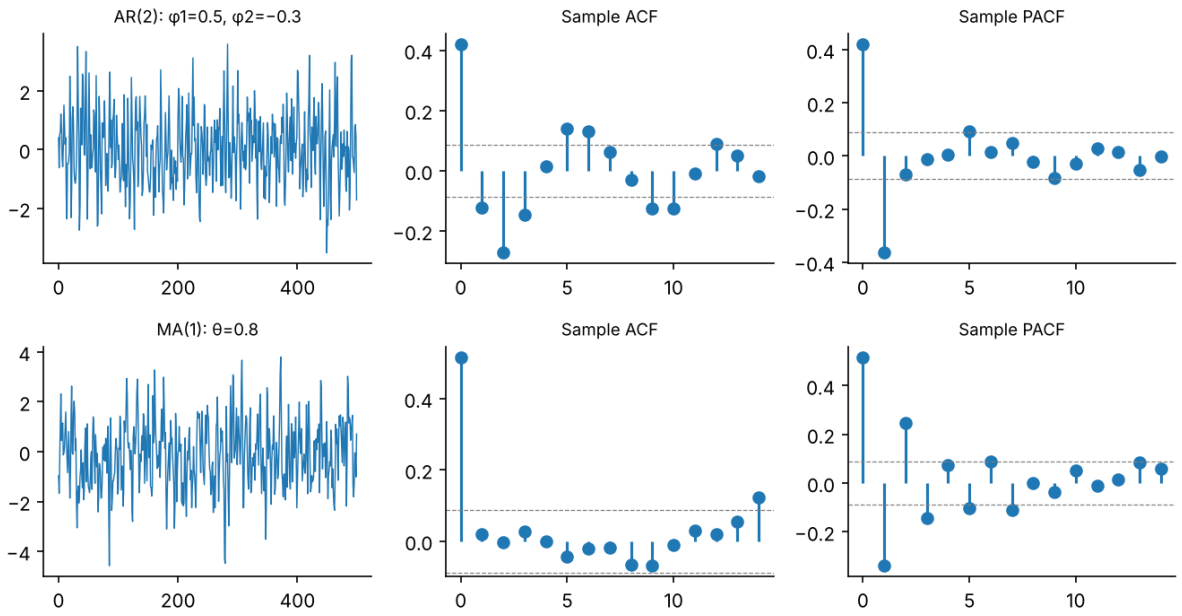
The Yule–Walker equations give $\rho_1 = \phi_1/(1-\phi_2)$ and $\rho_k = \phi_1\rho_{k-1} + \phi_2\rho_{k-2}$ for $k \geq 2$.

Checked: AR(2) with coefficients 0.5 and 0.3

Yule–Walker: $\rho_1 = 0.5/0.7 = 0.7143$ and $\rho_2 = 0.5 \times 0.7143 + 0.3 = 0.6571$. `statsmodels` gives the same values. I also tested five pairs (ϕ_1, ϕ_2) by computing the roots: $(0.5; 0.3)$ is stationary; $(1.2; -0.2)$, $(0.5; 0.6)$, $(1.0; 0)$ and $(-0.5; 0.5)$ are not, which matches the triangle.

7.5 Signatures in the ACF and PACF

Model	ACF	PACF
AR(p)	decays gradually	**cuts off** after p
MA(q)	**cuts off** after q	decays gradually
ARMA(p, q)	decays	decays



Simulated series (500 points) and their sample ACF and PACF. Top row: AR(2) ($\phi_1 = 0.5$, $\phi_2 = -0.3$): the PACF cuts off at 2. Bottom row: MA(1) ($\theta = 0.8$): the ACF cuts off at 1. Dotted lines are the $\pm 1.96/\sqrt{n}$ bands.

Pitfall

The $\pm 1.96/\sqrt{n}$ bands hold under the white-noise hypothesis. For a series that is *not* white noise, the correct (Bartlett) bands are wider, and in short samples the sample ACF is biased downward. Interpret the plots as a clue, not a verdict.

Exercise 7.1. In an AR(1) with $\phi = 0.6$ and $\sigma^2 = 1$, compute γ_0, ρ_1 and ρ_3.

Solution. $\gamma_0 = \sigma^2/(1-\phi^2) = 1/0.64 = 1.5625$. $\rho_1 = 0.6$ and $\rho_3 = 0.6^3 = 0.216$.

Exercise 7.2. An MA(1) has $\theta = -0.4$ (positive convention). What is ρ_1? And if the book's convention were the negative one, with the same number -0.4 printed?

Solution. Positive convention: $\rho_1 = -0.4/(1+0.16) = -0.345$. Negative convention, with $\theta = -0.4$: $\rho_1 = -\theta/(1+\theta^2) = +0.345$. The same number with different conventions gives correlations of opposite signs.

Chapter 8

ARMA, ARIMA and the Box–Jenkins methodology

8.1 The ARMA model and the lag operator

With $By_t = y_{t-1}$, an ARMA(p, q) is written

$$\phi(B)\, y_t = \theta(B)\, \varepsilon_t, \qquad \phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p, \quad \theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q.$$

Two technical requirements: **stationarity** (roots of $\phi(B)$ outside the unit circle) and **invertibility** (roots of $\theta(B)$ outside the unit circle). The latter ensures the shock ε_t can be recovered from past data and that the model is identified (we saw that θ and $1/\theta$ give the same ACF).

8.2 ARIMA: what the letter "I" does

If the series has a unit root (random walk, stochastic trend), difference it: $\Delta y_t = y_t - y_{t-1}$. An ARIMA$(p, d, q)$ is an ARMA(p, q) applied to the series differenced d times. $d > 2$ is almost never necessary.

Overdifferencing

Differencing a series that is already stationary creates a unit root in the MA component (the MA becomes noninvertible). Symptoms: sample ACF with ρ_1 close to -0.5 and a larger variance after differencing. Difference only as much as necessary.

8.3 The Box–Jenkins cycle in four steps

1. **Specification**: look at the plot of the series, decide the transformation (log for growing variance) and the degree of differencing (unit root test, chapter 10). Use the ACF and PACF to suggest candidates and information criteria (AIC, BIC) to compare them.
2. **Estimation**: maximum likelihood or conditional least squares.
3. **Diagnostics**: the residuals should look like white noise. Examine the ACF of the residuals and run the Ljung–Box test.
4. **Forecasting**: only with the validated model (chapter 9).

Pitfall: degrees of freedom of Ljung–Box

The statistic $Q = n(n+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(n-k)$ is compared with a chi-square with $m-p-q$ degrees of freedom, **not** m, when testing residuals of an estimated ARMA(p,q). Using m makes the test conservative. Many software packages let you enter $p+q$ to adjust. Check.

AIC versus BIC

Both penalize complexity: AIC $= -2\ln L + 2k$ and BIC $= -2\ln L + k\ln n$. The BIC penalizes more and tends to choose smaller, consistent models (it finds the true model when it is among the candidates). The AIC tends to favor forecast quality. When they disagree, look at both and prefer parsimony.

8.4 Seasonality

For data with a cycle of period s (12 in monthly data), the seasonal ARIMA $(p,d,q)(P,D,Q)_s$ is used: it adds seasonal differencing $(1-B^s)$ and AR and MA terms at lags $s, 2s, \ldots$ The classic example is the "airline model", an ARIMA$(0,1,1)(0,1,1)_{12}$ applied to the logarithm of the series. Seasonality can also be handled with indicator variables or with exponential smoothing (Holt–Winters).

Exercise 8.1. Is the model $y_t = 0.5y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}$ stationary and invertible?

Solution. AR part: $\phi(B) = 1 - 0.5B$, root $B = 2$, outside the unit circle: stationary. MA part: $\theta(B) = 1 + 0.4B$, root $B = -2.5$, outside the circle: invertible.

Exercise 8.2. A series has a sample ACF that decays very slowly (0.95; 0.93; 0.91, …). What does this suggest and what is the next step?

Solution. Very persistent autocorrelations are typical of a unit root or a trend. The next step is to apply a unit root test (ADF, KPSS) and, if confirmed, difference the series before identifying the ARMA.

Chapter 9

Forecasting

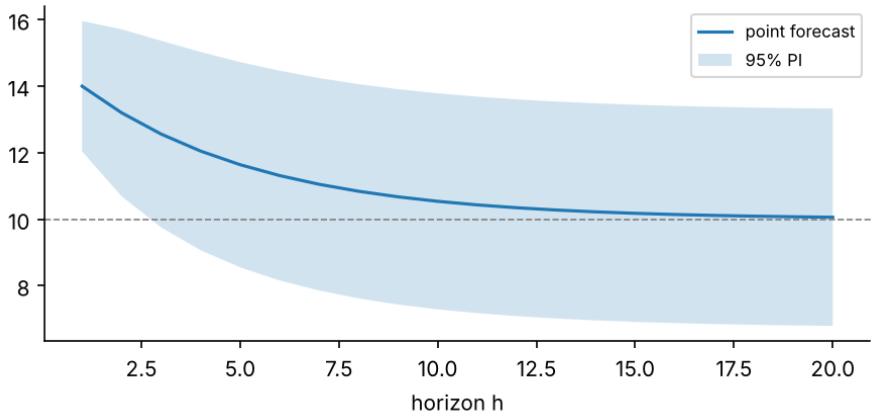
9.1 The best predictor

For squared error, the best predictor of y_{T+h} given the past is the conditional expectation $\mathbb{E}(y_{T+h} \mid y_T, y_{T-1}, \dots)$ (chapter 1). In an AR(1) with mean μ:

$$\hat{y}_{T+h} = \mu + \phi^h (y_T - \mu), \qquad \operatorname{Var}(e_{T+h}) = \sigma^2 \, \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

Intuition

The farther the horizon, the more the forecast approaches the long-run mean, and the more the uncertainty approaches the unconditional variance $\gamma_0 = \sigma^2/(1-\phi^2)$. For large h, the model knows no more than the mean.



AR(1) with $\mu = 10$, $\phi = 0.8$, $\sigma = 1$ and last deviation $y_T - \mu = 5$. The forecast decays to the mean and the interval widens and stabilizes.

Checked by simulation: interval coverage

I simulated 4000 AR(1) series with $\phi = 0.8$, forecast $h = 3$ steps ahead and built the 95% interval. The empirical coverage was 95.0%. The theoretical variance of the error for $h = 3$ is $(1 - 0.8^6)/(1 - 0.64) = 2.0496$, confirmed.

9.2 Random walk and ARIMA

In a random walk, $\hat{y}_{T+h} = y_T$ (the "naive" forecast) and $\operatorname{Var}(e_{T+h}) = h\sigma^2$. The interval grows without bound, which reflects the nature of permanent shocks. ARIMA models with $d \geq 1$

have the same property, which is very different from a stationary model.

Pitfall: optimistic intervals

Classical intervals treat the estimated parameters as true and assume the model is right. With short samples or model choice made on the same data, the real coverage is usually lower than the nominal one. In serious forecasting, evaluate coverage out of sample.

9.3 How to evaluate forecasts

- **Out of sample, always.** In-sample fit rewards complicated models.
- **Rolling origin**: estimate up to t, forecast $t+1, \dots, t+h$, move the origin forward and repeat.
- **Metrics**: MAE and RMSE (same unit as the series); MAPE is intuitive, but explodes if there are values near zero and penalizes positive and negative errors asymmetrically; MASE compares with the naive forecaster and is comparable across series.
- **Always include a benchmark**: naive (last value) or seasonal naive. A model that cannot beat the naive one is not worth using.

Exercise 9.1. AR(1) with $\mu = 0$, $\phi = 0.8$, $\sigma^2 = 1$ and last value $y_T = 5$. Give the forecast for $h = 3$ and the 95% interval.

Solution. Forecast: $0.8^3 \times 5 = 0.512 \times 5 = 2.56$. Error variance: 2.0496, standard deviation 1.4317. Interval: $2.56 \pm 1.96 \times 1.4317 = 2.56 \pm 2.806$, that is, $[-0.25;\ 5.37]$.

Exercise 9.2. A random walk has $\sigma = 2$. What is the standard deviation of the 9-step-ahead forecast error?

Solution. $\sqrt{h}\,\sigma = \sqrt{9} \times 2 = 6$.

Chapter 10

Unit roots, spurious regression and cointegration

10.1 Deterministic versus stochastic trend

Two series can "go up" for very different reasons.

- **Deterministic trend**: $y_t = a + bt +$ (stationary). Deviations from the line are temporary. Removing the trend makes the series stationary.
- **Stochastic trend** (unit root): $y_t = y_{t-1} + \varepsilon_t$. Shocks are permanent. Removing a line does *not* solve it; you must difference.

Treating one as if it were the other leads to serious forecasting and inference errors.

10.2 The augmented Dickey–Fuller (ADF) test

We estimate

$$\Delta y_t = \alpha + \gamma\, y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta y_{t-j} + e_t,$$

and test $H_0 : \gamma = 0$ (there is a unit root) against $\gamma < 0$ (stationary). The extra lags serve to whiten the residuals.

ADF pitfalls

- **The t statistic follows neither the t nor the normal distribution**. The critical values are the Dickey–Fuller ones, more negative (for example, about -2.86 at 5% with a constant). Comparing with -1.96 rejects too often.
- **The deterministic terms matter**: without constant, with constant or with trend, the critical values change and the wrong choice distorts the test.
- **Low power**: the test has difficulty telling a unit root from a root close to 1 (for example, $\phi = 0.98$) in moderate samples.
- **KPSS reverses the null hypothesis** (the null is stationarity). Using both together is common practice: ADF rejects and KPSS does not reject $\Rightarrow$ evidence in favor of stationarity.

Checked by simulation

In a 300-point random walk, the ADF p-value was 0.50 (does not reject the unit root, correct). In white noise, the p-value was $\approx 10^{-29}$ (strongly rejects, correct).

10.3 Spurious regression

Regressing an integrated series on another, independent of it, produces "significant" coefficients with absurd frequency.

Checked by simulation

I simulated 1000 pairs of *independent* random walks with 200 observations each, ran OLS of one on the other and tested the slope at 5%. The t test rejected the null of zero slope in **83.6%** of the cases, instead of the expected 5%.



Left: five independent random walks, which seem to have trends of their own. Right: two independent series that, by chance, look related.

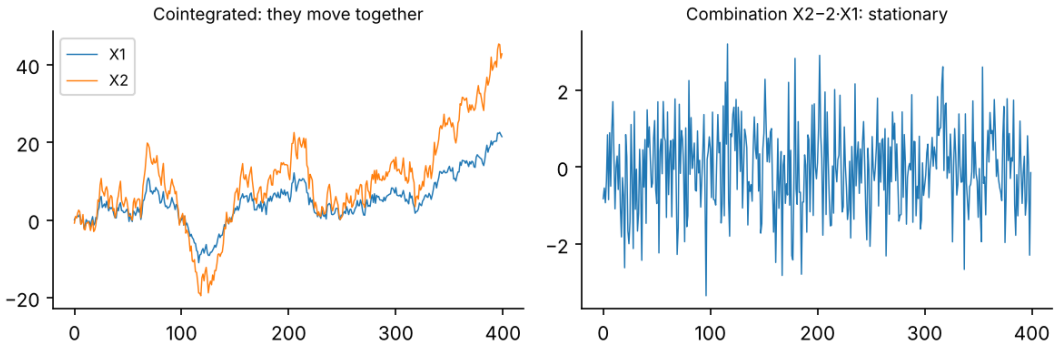
Practical rules: (i) never run OLS on levels of series with a unit root without investigating cointegration; (ii) a classic warning sign is a high R^2 with strongly autocorrelated residuals (Durbin–Watson statistic close to 0); (iii) if there is no cointegration, work with the differenced series.

10.4 Cointegration

Two $I(1)$ series are **cointegrated** if some linear combination of them is stationary. They can drift apart in the short run, but a force brings them back (a long-run equilibrium).

The drunk and the dog

Two independent random walks are like a drunk and a dog on the loose: each goes wherever it wants. Cointegration is the drunk with the dog on a leash: both wander aimlessly, but the distance between them is bounded.



$X_1 = w + \text{noise}$ and $X_2 = 2w + \text{noise}$, with w a random walk. Each series is nonstationary, but $X_2 - 2X_1$ is stationary.

Engle–Granger test (two steps): (1) regress one series on the other; (2) apply ADF to the residuals, using *specific* critical values for estimated residuals (more demanding than those of the ordinary ADF). The multivariate alternative is the Johansen test, which identifies how many cointegrating relations exist.

Checked by simulation

With 200 observations, I repeated 500 times. For cointegrated pairs, the Engle–Granger test at 5% rejected "no cointegration" in 99.6% of the simulations. For independent pairs, it rejected in 7.4% (slightly above the nominal 5% in moderate samples, but very far from 84%).

Error correction model: if y_t and x_t are cointegrated, $\Delta y_t = \alpha(y_{t-1} - \beta x_{t-1}) + \gamma \Delta x_t + e_t$, with $\alpha < 0$ measuring the speed at which the economy returns to equilibrium.

Exercise 10.1. A regression between two price series in levels gives $R^2 = 0.95$ and Durbin–Watson of 0.1. What should be done before trusting this result?

Solution. The pattern (high R^2, very low DW) is typical of spurious regression. Test for a unit root in each series; if both are $I(1)$, test for cointegration (Engle–Granger or Johansen). Without cointegration, the regression in levels has no interpretation and the correct approach is to model the differences. With cointegration, use an error correction model.

Chapter 11

Vector autoregressions (VAR)

11.1 The model

A VAR(p) for a vector of k variables is

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}(u_t) = 0, \quad \mathbb{E}(u_t u_t') = \Sigma.$$

Each variable is explained by the lags of *all* variables, including its own. Estimation by OLS, equation by equation, is consistent and, when all equations have the same regressors, is equivalent to maximum likelihood estimation (under normality).

Stability (stationarity): all eigenvalues of the companion matrix must have modulus less than 1. This is the multivariate analogue of the condition $|\phi| < 1$.

11.2 Choosing the number of lags

Estimate VARs with $p = 1, 2, \ldots, p_{\max}$ and compare AIC, BIC (or Hannan–Quinn). Check that the residuals have no autocorrelation.

11.3 Granger causality

x *Granger-causes* y if the lags of x help predict y beyond the lags of y itself. It is tested with a joint F test of the coefficients of x in the equation for y.

Pitfall: Granger is not causality

It is a notion of *predictive precedence*. If an omitted third factor moves both variables with different lags, "Granger causality" appears with no cause at all. The name is misleading.

Checked by simulation

I simulated a bivariate VAR(1) with $A = \begin{pmatrix} 0.5 & 0.2 \\ 0 & 0.4 \end{pmatrix}$ ($T = 1500$). Estimate: $\hat{A} = \begin{pmatrix} 0.51 & 0.195 \\ -0.005 & 0.399 \end{pmatrix}$. The eigenvalues of A have moduli 0.5 and 0.4 (stable). The Granger test correctly identified: y_2 helps predict y_1 (p-value $\approx 10^{-15}$) and y_1 does not help predict y_2 (p-value 0.82).

11.4 Impulse response functions and variance decomposition

The **impulse response** shows how each variable reacts, over time, to a shock in one equation. The problem: the errors u_t are usually correlated across equations, so "a shock only in y_1" does not exist without an additional assumption. The most common solution is the Cholesky decomposition, which imposes a recursive order (the first variable affects the others in the same period, but not the reverse).

Pitfall

The order of the variables in the Cholesky decomposition changes the results when the errors are correlated. This is an identification choice, and it must be justified economically, not decided for convenience. The forecast error variance decomposition (FEVD) inherits the same problem.

Exercise 11.1. A VAR(1) has $A = \begin{pmatrix} 0.6 & 0.3 \\ 0.2 & 0.5 \end{pmatrix}$. Is it stable?

Solution. Eigenvalues: $\lambda^2 - (0.6 + 0.5)\lambda + (0.6 \cdot 0.5 - 0.3 \cdot 0.2) = \lambda^2 - 1.1\lambda + 0.24 = 0$. So $\lambda = (1.1 \pm \sqrt{1.21 - 0.96})/2 = (1.1 \pm 0.5)/2$, that is, 0.8 and 0.3. Both have modulus less than 1: the VAR is stable.

Chapter 12

Volatility: ARCH and GARCH

12.1 The problem

Financial returns have three stable features: (i) almost no autocorrelation, (ii) strongly autocorrelated *squares* of returns (volatility comes in waves: calm, turbulence), (iii) heavier tails than a normal. A classic ARMA assumes constant variance and captures none of this.

12.2 ARCH(1) and GARCH(1,1)

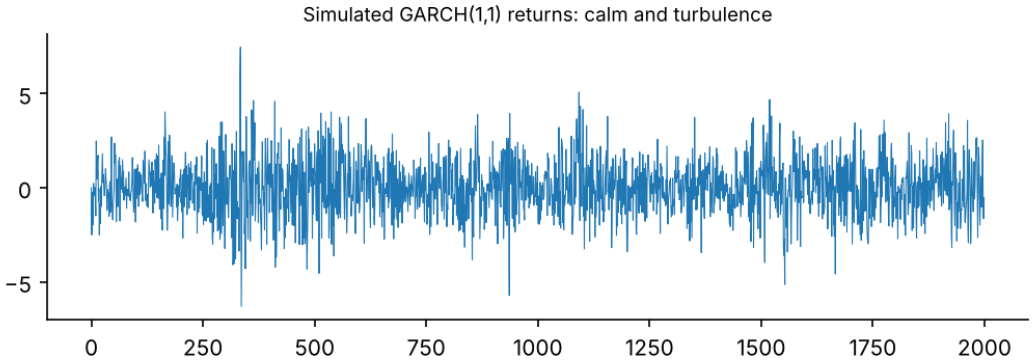
Write the (already demeaned) return as $r_t = \sigma_t z_t$, with z_t i.i.d. with mean 0 and variance 1. In the **GARCH(1,1)**:

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0.$$

- ARCH(1) is the case $\beta = 0$: today's volatility depends only on yesterday's shock.
- **Stationarity**: $\alpha + \beta < 1$. Unconditional variance: $\omega/(1 - \alpha - \beta)$.
- $\alpha + \beta$ measures **persistence**. Values close to 1 (such as 0.97) are typical in real data.
- The kurtosis (of r_t) is greater than 3 even with normal z_t.

Checked by simulation

With $\omega = 0.1$, $\alpha = 0.1$, $\beta = 0.85$ and a long simulation (400 thousand points): variance 2.009 (theory: $0.1/0.05 = 2$); kurtosis 3.774 (theory: 3.774); autocorrelation of r_t^2 at lag 1: 0.177 (theory: 0.179). The autocorrelation of r_t was close to zero (0.04 in a sample of 2000 points), but that of r_t^2 was 0.15. White noise in the returns, strong dependence in the volatility.



Estimation: individual parameters are hard

Fitting a GARCH(1,1) to 2000 simulated points, I obtained $\hat{\omega} = 0.054$, $\hat{\alpha} = 0.086$ and $\hat{\beta} = 0.890$, against the true 0.1; 0.1; 0.85. The individual parameters missed by quite a bit, but the combination that matters came out close: estimated persistence 0.976 (true 0.95) and unconditional variance $0.054/0.024 \approx 2.26$ (true 2.0). In very persistent models, ω, α and β become strongly correlated in estimation.

Pitfall

- **Do not apply ARCH to a series with a unit root or trend**; work with returns (differences of logarithms).
- **Symmetric GARCH does not distinguish good news from bad.** For the leverage effect (drops raise volatility more than rises), use GJR-GARCH or EGARCH.
- **Choosing normal z_t understates tails**; many use Student's t.

Exercise 12.1. GARCH(1,1) with $\omega = 0.2$, $\alpha = 0.1$, $\beta = 0.8$. Compute the unconditional variance and the forecast of σ^2_{t+1} given $r^2_t = 4$ and $\sigma^2_t = 3$.

Solution. Unconditional variance: $0.2/(1 - 0.9) = 2$. Forecast:
$\sigma^2_{t+1} = 0.2 + 0.1 \times 4 + 0.8 \times 3 = 0.2 + 0.4 + 2.4 = 3.0$.

Exercise 12.2. Why does $\alpha + \beta = 1$ create problems, and what is this case called?

Solution. With $\alpha + \beta = 1$ the unconditional variance does not exist (the denominator is zero) and volatility shocks never dissipate. This is the IGARCH. The forecast of future volatility no longer reverts to a long-run mean.

Chapter 13

Machine learning for time series

13.1 What changes when data have an order

Machine learning methods (random forests, boosting, neural networks) were designed, for the most part, for independent observations. In time series, three precautions are mandatory.

13.1.1 Cross-validation that respects time

Random cross-validation (shuffled *k-fold*) mixes past and future. Since neighboring observations are similar, the model "sees" points almost identical to the one it is going to predict. This leaks information and understates the error.

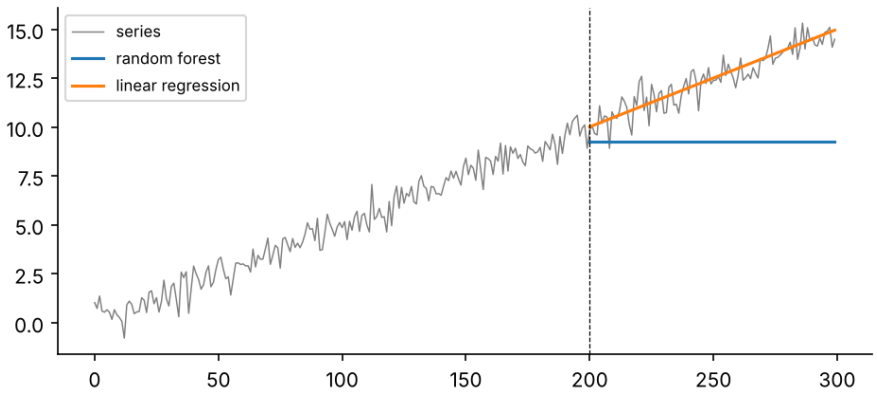
Checked by simulation: leakage in validation

A very persistent AR(1) series ($\phi = 0.98$, noise with standard deviation 1), random forest with 5 lags as explanatory variables. RMSE with random cross-validation: 1.08. RMSE with temporal validation (training on the past, testing on the future): 1.49. The irreducible noise is 1.0. The random procedure looks almost perfect. The temporal one shows the real performance, much worse. The difference is pure leakage.

Always use **rolling or expanding window validation** (rolling origin, as in chapter 9) and, when windows overlap, leave a gap between training and test.

13.1.2 Trees do not extrapolate

A tree (and a forest) predicts averages of observations already seen. If the series has a trend and the future leaves the training range, it gets "stuck" at the last level.



A series with a linear trend. The random forest, trained up to $t = 200$, predicts a flat level; linear regression continues the trend.

Checked by simulation

At the point $t = 299$, the forest predicted 9.27, and the value of the trend is 14.95.

Solutions: difference or detrend before fitting, model the change instead of the level, or use a hybrid model (linear trend plus forest on the residuals).

13.1.3 Building the explanatory variables

To turn forecasting into a regression problem: use lags $(y_{t-1}, \dots, y_{t-p})$, moving averages, calendar variables (month, day of the week, holidays) and external regressors. Any statistic built with future data (for example, normalizing with the mean of the whole series) is leakage.

13.2 Two useful families

- **Global models**: instead of fitting one model per series, train a single model (for example, a recurrent neural network or boosting) with thousands of series. It helps when each series is short or noisy. It is the central idea of the Faloutsos et al. tutorial on large-scale forecasting.
- **Time series classification**: the problem is not to forecast, but to assign a label to an entire series (for example, normal versus anomaly in an electrocardiogram). It is the focus of the `pyts` package, which offers techniques such as dynamic time warping, shapelets and image representations. Do not confuse it with forecasting.

Pitfalls

- **Always compare with the naive forecaster and with an ARIMA/ETS**. Many dazzling deep-network results disappear in front of a simple benchmark.
- **Prediction intervals**: ML models give point forecasts. For uncertainty, use quantile regression, conformal prediction or simulation.
- **Hyperparameters tuned on the test data**: the test set must be touched only once.

Exercise 13.1. Why is shuffling the observations before splitting into training and test a mistake in a time series?

Solution. Because it makes the model train on observations from the future and test on the past, and because adjacent observations are strongly dependent: test points have near-duplicates in training. The estimated error becomes unrealistically low. The correct test respects chronological order.

Part V

Quick reference

Chapter 14

Pitfalls and conventions cheat sheet

This list gathers the points where books, software and intuition tend to diverge. Each item appears in the chapter indicated.

#	Pitfall	Correct practice
1	Taking $\mathbb{E}(u \mid x) = 0$ as necessary for OLS consistency (ch. 3)	$\mathbb{E}(x'u) = 0$ is enough. The conditional assumption is needed for the conditional-expectation interpretation and for finite-sample results.
2	Correcting for heteroskedasticity only after "testing" (ch. 3)	Report robust errors directly: they are valid in both cases.
3	Forgetting that heteroskedasticity-robust errors do not correct autocorrelation (ch. 3, 6)	HAC in series, clustered errors in panels.
4	Interpreting omitted variable bias without checking both signs (ch. 4)	The bias is $\beta_2\delta_1$. The sign is the product of the two.
5	Expecting measurement error to always attenuate (ch. 4)	Only holds for classical error in one regressor. With several regressors or nonclassical error, the effect is uncertain.
6	Running the 2nd stage of IV by hand (ch. 5)	Use a 2SLS command, which corrects the standard errors.
7	Trusting the "F $>$ 10" rule as a guarantee (ch. 5)	It is a minimum. Also report weak-instrument-robust intervals.
8	Using classic Hausman with robust errors (ch. 6)	Use the Mundlak version with clustered errors.
9	Forgetting that fixed effects require strict exogeneity (ch. 6)	With a lagged dependent variable or *feedback*, use dynamic panel (GMM).
10	Copying an MA formula from a book and a coefficient from software without checking the sign (ch. 7)	Cryer–Chan and Mills: $\varepsilon_t - \theta\varepsilon_{t-1}$. Hamilton, R, `statsmodels`: $\varepsilon_t + \theta\varepsilon_{t-1}$.
11	Using $\pm 1.96/\sqrt{n}$ bands as a verdict (ch. 7)	They hold under white noise. For other series, the bands are wider.
12	Ljung–Box with m degrees of freedom on ARMA residuals (ch. 8)	Use $m - p - q$.
13	Comparing the ADF statistic with -1.96 (ch. 10)	Use Dickey–Fuller critical values (about -2.86 at 5% with a constant).
14	Regressing integrated series in levels (ch. 10)	Test cointegration. If there is none, difference.
15	Assuming the prediction interval has exact coverage (ch. 9)	It ignores parameter and model uncertainty. Evaluate coverage out of sample.
16	Reading Granger as causality (ch. 11)	It is predictive precedence.
17	Ignoring the Cholesky order in impulse responses (ch. 11)	The order is an identification assumption and changes the results.

#	Pitfall	Correct practice
18	Interpreting ω, α, β in isolation in persistent GARCH (ch. 12)	Look at the persistence $\alpha + \beta$ and the unconditional variance.
19	Random cross-validation on series (ch. 13)	Temporal validation, with rolling origin.
20	Expecting trees to extrapolate a trend (ch. 13)	Remove the trend or use a hybrid model.

Suggested study plan

1. Chapters 1 to 3: do all the exercises by hand.
2. Chapters 4 and 5: replicate the simulations in Python or R, changing the parameters, until you can predict the result before running.
3. Chapter 6: build a simulated panel with c_i correlated and uncorrelated and compare the estimators.
4. Chapters 7 to 9: fit an ARIMA to a real series, run the diagnostics and the rolling-origin evaluation.
5. Chapters 10 to 12: test for unit roots and cointegration in pairs of price series; estimate a GARCH on daily returns.
6. Chapter 13: reproduce the cross-validation leakage and then solve it with temporal validation.